

Problem Set for Integer Programming

1. A typical large oil and gas company operates many explorations and production projects, which involve several billion dollars every year. These companies are annually faced with the problem of where the capital should be spent and which combination of projects should be selected from several possible project mixes. They have the difficult task of portfolio selection from a large number of competing projects for immediate or future operation under a limited amount of investments.

Consider a small firm with 4 competing oil production projects. Table 1 presents the production, capital, and the net present value for these projects. They contact you to help select the best combination of projects under a certain amount of investment, while fulfilling the firm's goals.

Table 1: Portfolio optimization problem Data

Project	NPV (M\$)	Capital Required (M\$)	Production (Mbbl)
A	25	11	28
B	20	9	20
C	19	14	25
D	28	17	30

- (a) Formulate an integer program to maximize the net present value (NPV) subject to a capital limit stating the firm can spend no more than \$32 million (M\$) and a production level stating the firm must produce at least 73 million barrels (Mbbl).
- (b) Suppose that there are the following additional constraints:
 - (i) If Project A is selected, then Project B is also selected;
 - (ii) Either Project A is selected or Project C selected, but not both;
 - (iii) At least one of Projects A, B, and D is selected.

Extend your integer program to satisfy these constraints.

2. We are given a mixed integer program defined as follows:

$$\begin{aligned}
 \max \quad & 10x_1 + 22x_2 + 5x_3 + 15x_4 + 17x_5 + 12x_6 + 4x_7 \\
 \text{s.t.:} \quad & 5x_1 + 3x_2 + 8x_3 + 9x_4 + 16x_5 + 5x_6 + 10x_7 \leq 700 \\
 & x_i \in \{0, 1\}, \quad \text{for } i = 1, 2, 3 \\
 & 0 \leq x_i \leq 200, \quad \text{for } i = 4, 5, 6, 7
 \end{aligned}$$

For each of the parts below, you are to add constraint(s) and possibly variables to ensure that the logical condition is satisfied by the integer program. Each part is independent; that is, no part depends on the parts preceding it. You do not need to repeat the integer programming objective or constraints given above. You may use the big M method for formulating constraint when it is appropriate.

- (a) Write a single linear constraint that is equivalent to the statement "If $x_2 = 1$ is selected, then $x_1 = 0$ "

- (b) Write a single linear constraint that is equivalent to the statement “ x_1 and x_3 cannot both be 1”.
 - (c) Add 3 binary variables w_1, w_2 , and w_3 and at most 4 constraints so as to ensure that at least two of the following constraints is satisfied: (i) $x_4 \geq 50$, (ii) $x_5 \leq 25$, (iii) $x_6 + x_7 \leq 100$.
 - (d) Add variable(s) and constraint(s) that ensure either $2x_4 + x_5 \leq 50$ or $4x_4 - x_5 \geq 20$, but not both.
 - (e) Add three binary variables w_5, w_6 , and w_7 and two constraints that ensures that $x_5 = 9$ or 15 or 20.
3. A local radio station is going to schedule 6 commercials (i.e., short audio advertisements) into 5 time slots. The lengths of the six commercials are 12, 18, 22, 35, 40, and 59 seconds, respectively. Each of the 5 time slots has 60 seconds. Can the radio station insert all commercials into the 5 slots? What is the smallest number of time slots used? Write down the integer programming model for this problem.
4. We have two binary variables $x_1, x_2 \in \{0, 1\}$. We want to represent the outcome of the two logical operations AND, OR, applied on x_1 and x_2 . The definition of these two operations is as follows:
- Let $w = (x_1 \text{ AND } x_2)$. w is 1 if and only if both x_1 and x_2 are 1, and is 0 otherwise.
 - Let $y = (x_1 \text{ OR } x_2)$. y is 1 if and only if at least one of the variables x_1 and x_2 is 1, and is 0 otherwise.

Your goal in this problem is to define w, y using linear constraints only. For the AND and OR operations, you are not allowed to introduce additional variables.

- (a) Write a set of linear constraints that define $w = (x_1 \text{ AND } x_2)$. The constraints should only involve the variables w, x_1, x_2 .
- (b) Write a set of linear constraints that define $y = (x_1 \text{ OR } x_2)$. The constraints should only involve the variables y, x_1, x_2 .

Solution

1. (a) We define four binary variables x_A, x_B, x_C, x_D as follows:

$$x_A := \begin{cases} 1 & \text{if Project A is selected,} \\ 0 & \text{otherwise.} \end{cases} \quad x_B := \begin{cases} 1 & \text{if Project B is selected,} \\ 0 & \text{otherwise.} \end{cases}$$

$$x_C := \begin{cases} 1 & \text{if Project C is selected,} \\ 0 & \text{otherwise.} \end{cases} \quad x_D := \begin{cases} 1 & \text{if Project D is selected,} \\ 0 & \text{otherwise.} \end{cases}$$

Then our formulation is:

$$\begin{aligned} \max \quad & 25x_A + 20x_B + 19x_C + 28x_D \\ \text{s.t.} \quad & 11x_A + 9x_B + 14x_C + 17x_D \leq 32, \\ & 28x_A + 20x_B + 25x_C + 30x_D \geq 73, \\ & x_A, x_B, x_C, x_D \in \{0, 1\}. \end{aligned}$$

- (b) The additional constraints for each part are as follows:

- (i) $x_A \leq x_B$ or equivalently $x_A - x_B \leq 0$;
- (ii) $x_A + x_C = 1$;
- (iii) $x_A + x_B + x_D \geq 1$.

2. (a) $x_1 + x_2 \leq 1$
 (b) $x_1 + x_3 \leq 1$
 (c)

$$\begin{aligned} x_4 &\geq 50 - M(1 - w_1), \\ x_5 &\leq 25 + M(1 - w_2), \\ x_6 + x_7 &\leq 100 + M(1 - w_3), \\ w_1 + w_2 + w_3 &\geq 2. \end{aligned}$$

- (d)

$$\begin{aligned} 2x_4 + x_5 &\leq 50 + M(1 - w_8), \\ 4x_4 - x_5 &\geq 20 - M(1 - w_9), \\ w_8 + w_9 &= 1. \end{aligned}$$

- (e) $x_5 = 9w_5 + 15w_6 + 20w_7$ and $w_5 + w_6 + w_7 = 1$

3. Suppose we let $y_j = 1$ if block j is used and $y_j = 0$ otherwise and $x_{ij} = 1$ if commercial i is assigned to block j and $x_{ij} = 0$ otherwise. Based on this choice of decision variables,

the formulation is as follows:

$$\begin{aligned}
\min \quad & y_1 + y_2 + y_3 + y_4 + y_5 \\
\text{s.t.} \quad & 12x_{1j} + 18x_{2j} + 22x_{3j} + 35x_{4j} + 40x_{5j} + 59x_{6j} \leq 60y_j, \quad \text{for } j = 1, \dots, 5 \\
& x_{i1} + x_{i2} + x_{i3} + x_{i4} + x_{i5} = 1, \quad \text{for } i = 1, \dots, 6 \\
& x_{ij}, y_j \in \{0, 1\} \quad \text{for } j = 1, \dots, 5 \text{ and } i = 1, \dots, 6
\end{aligned}$$

The first set of constraints represent the condition that each possible arrangement contains at most 60 seconds. The second set of constraints represent the condition that each commercial is assigned to exactly one time slot. We then minimize the total number of blocks. Additionally, we need all of our decision variables to be binary.

By inspection, however, the minimum number of blocks needed is 4; the sum of all commercial times is 186, meaning that more than 3 are needed, and there is a clear way to use only 4.

4. (a) There are multiple acceptable formulations; one is the following:

$$\begin{aligned}
w &\leq x_1 \\
w &\leq x_2 \\
w &\geq x_1 + x_2 - 1
\end{aligned}$$

- (b) Again, there are multiple acceptable formulations; one is the following:

$$\begin{aligned}
y &\geq x_1 \\
y &\geq x_2 \\
y &\leq x_1 + x_2
\end{aligned}$$